AI Coaching for Lesson Input Fields

CSE 360 Team Project Phase Two, Team 53. Team research compiled and integrated on 2026-10-28. Field names are the stored attribute names in Member 1's schema (Vishwam Gupta). Assignments follow the supplied October 28 team compilation. This is planning research for the Phase Three AI epics; no AI feature is claimed as implemented in TP2.

The team's HW2 evaluation of the course-provided LessonShare v3a supplies the starting points: brainstorming questions, writing feedback, summarization, keyword suggestions and takeaway extraction [S15]. The external tools below are proposed alternatives or complements to those coaching functions; each suggestion still requires human acceptance.

5.2 Prioritized field list

Priority 1 is the field where coaching most directly improves reuse. Fields are ordered as agreed in the supplied team compilation: lesson, situation and title first, followed by experience actions, search support, classification, effort, duration and private review.

#	Field (stored name)	Table	Owner	Story / epic	Why this priority
1	**lessonText** Lesson Text	lessonDB	Joshua Luther	Contributor C1; Contributor Phase 3 epic	The reusable takeaway; the field future readers rely on most.
2	**situation** Situation	lessonDB	Kanish Garg	Contributor C1; Contributor Phase 3 epic	Context tells a later reader when the lesson applies.
3	**title** Title	lessonDB	Joshua Luther	Contributor C1-C2; Viewer search (Phase 3)	Titles drive the My Lessons list and later searches.
4	**whatDone** What was done (experience)	lessonExperienceDB	Vishwam Gupta	Contributor C3	Records the new attempt to apply the lesson.
5	**howDone** How it was done (experience)	lessonExperienceDB	Vishwam Gupta	Contributor C3	Steps and tools make the attempt repeatable.
6	**searchTerms** Search terms (curator only)	lessonCuratorDB	Aditya Gupta (agupt545)	Curator epic: AI-built search terms	Lets users find a lesson when their words differ from the lesson's words.
7	**category** Category	lessonDB	Kanish Garg	Contributor C1; Viewer search (Phase 3)	A consistent category groups lessons for browsing and search.
8	**effortHours** Member effort hours (memberName, effortHours)	lessonEffortDB	Aditya Gupta (agupt515)	Contributor C3	Per-member effort is the raw data for the Curator time-and-effort epic.
9	**durationHours** Duration in hours (experience)	lessonExperienceDB	Aditya Gupta (agupt515)	Contributor C3	Elapsed time, separate from effort, shows how long reuse really takes.
10	**notes** Curator notes (curator only)	lessonCuratorDB	Aditya Gupta (agupt545)	Contributor C1 private curator information; Curator Phase 3 refinement	Records private review context without exposing it to Contributors.

5.3 Field assignments (two per member)

Member	Fields researched	Section source
Kanish Garg	situation (#2), category (#7)	Supplied team AICoaching compilation, reviewed 2026-10-28

Member	Fields researched	Section source
Vishwam Gupta	whatDone (#4), howDone (#5)	Supplied team AICoaching compilation, reviewed 2026-10-28
Joshua Luther	lessonText (#1), title (#3)	Supplied team AICoaching compilation, reviewed 2026-10-28
Aditya Gupta (agupt515)	effortHours (#8), durationHours (#9)	Supplied team AICoaching compilation, reviewed 2026-10-28
Aditya Gupta (agupt545)	searchTerms (#6), notes (#10)	Supplied team AICoaching compilation, reviewed 2026-10-28

Rules for every field

AI output is a suggestion: the author (or Curator) must accept it, nothing is saved automatically, and a rejected suggestion leaves the original text unchanged. Suggestions never invent facts, events, numbers or results. Locked fields and locked lessons receive no suggestions. Every accepted value must still pass the existing validation (LessonRecognizer or ExperienceRecognizer). If the AI service is unavailable, the form keeps the typed draft. Evaluate each tool on non-confidential sample lessons before choosing it; free availability and limits must be rechecked before implementation. Application requests must use a self-hosted LanguageTool server: the free public endpoint prohibits automated requests. Total person-hours can exceed elapsed time; four people working three hours legitimately produce twelve person-hours. AI may ask about implausible entries but must not require those totals to match.

5.4 Coaching requirements, free AI tools and citations

1. lessonText - Lesson Text - owner: Joshua Luther

Priority	Coaching requirement	Acceptance check
P1	Ask focused questions when the action, outcome or practical takeaway is missing. Never invent events.	Vague input produces questions; no new facts.
P1	Flag ambiguous wording and offer a clearer version that keeps technical terms and stated uncertainty.	Every claim in a suggestion is in the input.
P2	Explain spelling, grammar and blaming-tone issues with specific fixes.	Each proposed edit has a reason.
P2	Require explicit acceptance; keep the original if rejected.	Nothing is saved automatically.
P3	Keep suggestions within the 20-2000 character rule.	LessonRecognizer accepts any accepted suggestion.

Free AI tool: A local Ollama model with schema-constrained feedback for clarity, completeness and alternatives [S4][S7]; self-hosted LanguageTool for spelling/grammar [S5][S6]. Each model license must be checked separately.

Citations: [S4] https://ollama.com/blog/structured-outputs; [S7] https://github.com/ollama/ollama/blob/main/LICENSE; [S5] https://dev.languagetool.org/http-server; [S6] https://github.com/languagetool-org/languagetool

2. situation - Situation - owner: Kanish Garg

Priority	Coaching requirement	Acceptance check
P1	Ask for missing context: what the team was doing, when, and the impact. Never supply details the author did not give.	"It broke" prompts for what broke and its effect; no details are added.
P1	Flag text that states the lesson instead of the situation and suggest moving it to Lesson Text.	A takeaway sentence is flagged, not rewritten.
P2	Flag spelling, grammar and readability problems.	Each issue is shown with its position and a suggested fix.
P3	Respect the Situation field lock and the 20-2000 character rule.	A locked field offers no edit; suggestions stay within limits.

Free AI tool: LanguageTool run as your own local HTTP server (core is LGPL 2.1+) for grammar and style checks [S5][S6], and Ollama running a local model with schema-constrained output for the missing-context questions [S4].

Citations: [S4] https://ollama.com/blog/structured-outputs; [S5] https://dev.languagetool.org/http-server; [S6] https://github.com/languagetool-org/languagetool

3. title - Title - owner: Joshua Luther

Priority	Coaching requirement	Acceptance check
P1	Flag vague titles; offer alternatives grounded only in the Situation and Lesson Text.	No unsupported event or result appears.
P1	Flag a title that disagrees with the lesson and ask which is right.	An unrelated title is flagged, not silently changed.
P2	Offer up to three alternatives with reasons; the author may keep the original.	Nothing is saved automatically.
P2	Keep each suggestion within 5-100 characters and keep domain terms.	Every accepted title passes LessonRecognizer.

Free AI tool: A local Ollama model with schema-constrained feedback for clarity, completeness and alternatives [S4][S7]; self-hosted LanguageTool for spelling/grammar [S5][S6]. Each model license must be checked separately.

Citations: [S4] https://ollama.com/blog/structured-outputs; [S7] https://github.com/ollama/ollama/blob/main/LICENSE; [S5] https://dev.languagetool.org/http-server; [S6] https://github.com/languagetool-org/languagetool

4. whatDone - What was done (experience) - owner: Vishwam Gupta

Priority	Coaching requirement	Acceptance check
P1	Ask for a concrete action and observed outcome when absent. Never invent tasks, measurements or success.	"fixed it" prompts for the action and outcome.
P2	Suggest clearer, neutral wording.	Rewrites need explicit approval.
P3	Flag spelling and grammar.	Issues are listed, not auto-applied.

Free AI tool: Ollama with a local model returning structured feedback (missing details, questions, suggested revision) through a JSON schema [S4]. The Ollama runtime is MIT licensed; each model has its own license [S7].

Citations: [S4] https://ollama.com/blog/structured-outputs; [S7] https://github.com/ollama/ollama/blob/main/LICENSE

5. howDone - How it was done (experience) - owner: Vishwam Gupta

Priority	Coaching requirement	Acceptance check
P1	Coach for steps and tools, with enough detail to repeat the attempt.	"used the tool" prompts for the tool, version and steps.
P2	Ask for missing context without inventing commands.	No command or setting appears that the author did not give.
P3	Flag ambiguous wording and grammar; respect locks; keep the draft if the service fails.	A locked lesson offers no edit; a failure keeps the text.

Free AI tool: The same local Ollama structured-feedback approach [S4][S7], with a self-hosted LanguageTool server for proofreading [S5][S6].

Citations: [S4] https://ollama.com/blog/structured-outputs; [S5] https://dev.languagetool.org/http-server; [S6] https://github.com/languagetool-org/languagetool; [S7] https://github.com/ollama/ollama/blob/main/LICENSE

6. searchTerms - Search terms (curator only) - owner: Aditya Gupta (agupt545)

Priority	Coaching requirement	Acceptance check
P1	Propose key phrases taken from the lesson text and ranked by relevance; the Curator accepts or removes each one.	Every proposed phrase appears in, or is similar to, the lesson.
P1	Never show search terms to the Contributor (story C1).	Contributor views contain no search terms (testbed CR-08).
P2	Suggest related terms by meaning (for example "merge conflict" for "branches collided") and show the similarity score.	Each related term shows its score; low scores are not offered.
P3	Keep the list within the 2000-character column.	setCuratorData accepts the saved list.

Free AI tool: KeyBERT (MIT) for keyword and key-phrase extraction with BERT embeddings [S10], with Sentence Transformers (Apache-2.0) for meaning-based similarity [S11][S12].

Citations: [S10] https://github.com/MaartenGr/KeyBERT; [S11] https://www.sbert.net/; [S12] https://github.com/UKPLab/sentence-transformers

7. category - Category - owner: Kanish Garg

Priority	Coaching requirement	Acceptance check
P1	Suggest the best match from LessonsLearned.CATEGORIES only, with a confidence score, from the Title, Situation and Lesson Text.	The suggestion is always one of the six categories.
P1	Never change the category without the author choosing it.	The chosen value is the one saved.
P2	When the top two scores are close, show both and ask the author.	Ties are presented as a question.
P3	Respect the Category field lock.	A locked category offers no change.

Free AI tool: Zero-shot classification with the free MIT-licensed facebook/bart-large-mnli model through the Hugging Face Transformers pipeline, passing the six categories as candidate labels [S8][S9].

Citations: [S8] https://huggingface.co/facebook/bart-large-mnli; [S9] https://huggingface.co/tasks/zero-shot-classification

8. effortHours - Member effort hours (memberName, effortHours) - owner: Aditya Gupta (agupt515)

Priority	Coaching requirement	Acceptance check
P1	Turn typed amounts such as "90 min" or "half a day" into decimal hours and show the converted value for the author to confirm.	"90 min" is shown as 1.5 hours before saving.
P1	Explain person-hours separately from elapsed duration; ask only when an individual value is implausible for the described work, not merely because totals differ.	Four members working three hours can legitimately total 12 person-hours; no false mismatch.
P2	Ask about team members named in What or How who have no effort row.	Missing members are listed as a question.
P3	Keep the existing rules: each member once, non-negative hours, positive total.	ExperienceRecognizer accepts every accepted suggestion.

Free AI tool: A local Ollama model with a JSON schema (free; runtime MIT) for the consistency and missing-member questions [S4][S7], with Duckling (BSD), a rule-based parser rather than AI, turning text such as "3 mins" into structured durations [S13].

Citations: [S4] https://ollama.com/blog/structured-outputs; [S7] https://github.com/ollama/ollama/blob/main/LICENSE; [S13] https://github.com/facebook/duckling

9. durationHours - Duration in hours (experience) - owner: Aditya Gupta (agupt515)

Priority	Coaching requirement	Acceptance check
P1	Convert typed durations ("2 days", "45 minutes") to hours and show the result for confirmation.	"45 minutes" is shown as 0.75 hours.
P2	Explain the difference between elapsed duration and member effort when the author appears to confuse them. Ask whether a day means calendar time or working time.	No eight-hour or 24-hour assumption is silently applied; values need confirmation.
P3	Keep the positive, three-decimal limit.	ExperienceRecognizer accepts every accepted value.

Free AI tool: A local Ollama model with a JSON schema (free; runtime MIT) drafts the explanation and the confirmation question [S4][S7]. Duckling (BSD), a rule-based parser rather than AI, converts the typed amount to hours [S13].

Citations: [S4] https://ollama.com/blog/structured-outputs; [S7] https://github.com/ollama/ollama/blob/main/LICENSE; [S13] https://github.com/facebook/duckling

10. notes - Curator notes (curator only) - owner: Aditya Gupta (agupt545)

Priority	Coaching requirement	Acceptance check
P1	Ask what was changed or needs to change, and why; never invent a review decision.	A vague note produces a focused question, not an invented decision.
P1	Never show these notes or their coaching output to a Contributor.	Contributor projections and search results contain no private note text.
P2	Flag blaming language and offer constructive alternatives.	The Curator chooses whether to accept each revision.
P2	Flag spelling, grammar and accidental personal information.	Corrections are suggestions and private text stays on the local computer.

Free AI tool: A local Ollama model with structured output for tone and completeness [S4][S7], with a self-hosted LanguageTool server for spelling and grammar [S5][S6].

Citations: [S4] https://ollama.com/blog/structured-outputs; [S7] https://github.com/ollama/ollama/blob/main/LICENSE; [S5] https://dev.languagetool.org/http-server; [S6] https://github.com/languagetool-org/languagetool

Sources for proposed tools

[S4] Ollama, Structured outputs (2024-12-06). https://ollama.com/blog/structured-outputs

[S5] LanguageTool, Run your own HTTP server. https://dev.languagetool.org/http-server

[S6] LanguageTool source repository (LGPL 2.1 or later). https://github.com/languagetool-org/languagetool

[S7] Ollama runtime license (MIT). https://github.com/ollama/ollama/blob/main/LICENSE

[S8] Hugging Face, facebook/bart-large-mnli model card (MIT). https://huggingface.co/facebook/bart-large-mnli

[S9] Hugging Face, Zero-shot classification task guide. https://huggingface.co/tasks/zero-shot-classification

[S10] KeyBERT repository (MIT). https://github.com/MaartenGr/KeyBERT

[S11] Sentence Transformers documentation. https://www.sbert.net/

[S12] Sentence Transformers repository (Apache-2.0). https://github.com/UKPLab/sentence-transformers

[S13] Duckling repository (BSD). https://github.com/facebook/duckling

[S15] Course-provided LessonShare v3a: BrainstormService chat(), coachEntry(), extractTakeaways(), summarizeLesson() and suggestKeywords(); supplied team HW2 evaluation and October 28 coaching compilation. (Course-provided; no public URL required.)